Testing the AI's reasoning against the evidence

PRISM V2 · CASE-18016 (PLY-00230) · validation report, 2026-10-16

Synthetic data, not OLG customer data | AI output is provisional | A human investigator decides

BEFORE · live LLM run `INV-2f790604b4` (gpt-4o-mini)

- Lead assessment: **concern_reduced**
- F-13: the jackpot week explains the activity; the subject is "24.3% of the network"
- F-15: familiar retailers suggest ordinary behaviour
- F-14: "no information on historical behavior"
- Summary: "no definitive evidence of unusual behavior"

AFTER CHECKS · deterministic regression replay (no model calls)

- **F-13 rejected**: 24.3% is 9 of the subject's own 37 transactions
- **F-14 rejected**: the baseline holds 49 purchases over 23 days
- **Summary warning**: it denies anomalies that fired indicators record
- **concern_reduced flagged** for review; the label is not changed
- **11 of 13 findings accepted**; the original run is preserved

Fact (checked against the stored records)	Value	Evidence
Weekly spend vs. the subject's own baseline	$185.46 → $3,333.66 (**17.98×**)	`EV-001231`
Network-wide activity on event days	1,429.2 → 2,869.5 per day (**2.01×**)	`EV-001262`
Subject transactions on the jackpot days (Oct 10–11)	**9 of 37 = 24.3%**, all small ($5–$25)	`EV-001262`
Busiest 60 minutes	**12 × $250 prize claims**, Oct 12, 15:01–15:19, after the event	`EV-001258, EV-001225`
Historical baseline	49 purchases over 23 days, sufficient	`EV-001227`

Problem	Type	Detected now
24.3% read as a share of the network	Confirmed factual error	**Rejected**
History said to be missing despite a baseline	Confirmed factual error	**Rejected**
Jackpot week said to explain a burst that came after it	Unsupported interpretation	**Marked possible, not verified · peak outside event · downgraded (tested)**
Anomalies denied despite fired indicators	Unsupported interpretation	**Rejected / warned**
concern_reduced without covering counterevidence	Unsupported assessment	**Flagged, not overridden**
"Familiar retailers, so ordinary"; "purchases" for prize claims; new-retailer count	Unsupported or minor	**Not detected yet**

What the tests prove

- 5 regression tests built from this real run pass
- Agent tests 20/20; backend 114 passed, 1 timing-flaky test (passes on rerun)
- Evidence validator, tool permissions and detection rules unchanged

What remains untested

- No live AI run since the fix: we don't know the model's new conclusion
- No accuracy evaluation: no claim of improved fraud detection
- One synthetic case; the checks match wording patterns

Reproduce: `python scripts\validate_investigation.py INV-2f790604b4` (read-only) · `cd backend; python -m pytest -q tests/test_agents.py` (no API key). Full details: VALIDATION_REPORT.md